# Redwood Regional Bank CECL Review Memo

March 13, 2026

## Executive Summary

This memo documents an end-to-end CECL review of Redwood Regional Bank's Q1 2026 CECL Allowance Review. The supplied reserve engine was executable, the packaged baseline reserve was reproduced exactly, and scenario reruns plus targeted sensitivities were completed using the supplied portfolio, scenario, and overlay artifacts. Portfolio-level reserve direction was broadly reasonable with total ACL rising from $4,876,841 under Baseline to $7,132,022 under Adverse and $8,287,703 under Severe. The review nonetheless identified material exceptions requiring remediation before an unqualified approval posture would be appropriate.

## Scope and Coverage

This review assessed the supplied CECL package for internal coherence, quantitative reasonableness, and documentation alignment. The work covered baseline reproduction, scenario reruns, sensitivity testing, reserve driver analysis, and cross-checking of methodology, scenario, and overlay documentation against implemented behavior.

- Supported work: baseline reproduction, scenario reruns, sensitivity testing, driver analysis, and documentation cross-checking.
- Blocked work: none within the scope of the supplied package.
- Coverage note: conclusions are grounded in the supplied engine, portfolio, scenario files, overlays, and supporting documentation.

## Bank-Uploaded Input Package

- data/data_dictionary.csv (dataset)
- data/loan_level_snapshot.csv (dataset)
- data/overlay_schedule.csv (dataset)
- docs/control_process_note.md (document)
- docs/governance_minutes.md (document)

- docs/methodology.md (document)
- docs/model_overview.md (document)
- docs/overlay_memo.md (document)
- docs/prior_review_note.md (document)
- docs/scenario_assumptions.md (document)
- model/cecl_engine.py (code)
- outputs/prior_baseline_reserve.csv (packaged output table)
- outputs/prior_scenario_summary.csv (packaged output table)
- outputs/prior_segment_reserves.csv (packaged output table)
- scenarios/adverse.csv (scenario table)
- scenarios/baseline.csv (scenario table)
- scenarios/severe.csv (scenario table)

## Artifact Provenance and Review Record

This report uses explicit provenance markers so a reviewer can distinguish what the bank supplied from what Codex produced during the review. [BANK INPUT] indicates a file contained in the uploaded package under input_package/. [CODEX OUTPUT] indicates a review artifact created during discovery, planning, execution, or report rendering.

### Bank-Uploaded Artifacts

- **E001 [BANK INPUT]** `data/data_dictionary.csv` – dataset; CSV artifact with 13 data row(s); Generated as part of the Codex review record for this case.
- **E002 [BANK INPUT]** `data/loan_level_snapshot.csv` – dataset; CSV artifact with 1200 data row(s); Package inventory and case classification: Determine whether the upload supports a model-driven CECL review and identify the core evidence types available.; Baseline reproduction: Confirm that the packaged baseline reserve can be reproduced from the supplied engine and portfolio inputs.
- **E003 [BANK INPUT]** `data/overlay_schedule.csv` – dataset; CSV artifact with 12 data row(s); Overlay magnitude sensitivity: Assess the reserve effect of removing or amplifying qualitative overlays.
- **E004 [BANK INPUT]** `docs/control_process_note.md` – document; Text artifact with 3 line(s); Generated as part of the Codex review record for this case.
- **E005 [BANK INPUT]** `docs/governance_minutes.md` – document; Text artifact with 3 line(s); Generated as part of the Codex review record for this case.

- **E006** [**BANK INPUT**] `docs/methodology.md` – document; Text artifact with 20 line(s); Package inventory and case classification: Determine whether the upload supports a model-driven CECL review and identify the core evidence types available.; Forecast horizon sensitivity: Assess reserve sensitivity to changing the reasonable-and-supportable forecast horizon.; Reversion speed sensitivity: Assess how severe-scenario reserve responds to longer reversion periods.; Methodology and documentation alignment review: Test whether written methodology, model overview, and overlay documentation match implemented behavior and observed outputs.
- **E007** [**BANK INPUT**] `docs/model_overview.md` – document; Text artifact with 7 line(s); Methodology and documentation alignment review: Test whether written methodology, model overview, and overlay documentation match implemented behavior and observed outputs.
- **E008** [**BANK INPUT**] `docs/overlay_memo.md` – document; Text artifact with 3 line(s); Segment-level reasonableness review: Test whether scenario ordering remains directionally sensible at the segment level.; Overlay magnitude sensitivity: Assess the reserve effect of removing or amplifying qualitative overlays.; Methodology and documentation alignment review: Test whether written methodology, model overview, and overlay documentation match implemented behavior and observed outputs.
- **E009** [**BANK INPUT**] `docs/prior_review_note.md` – document; Text artifact with 3 line(s); Generated as part of the Codex review record for this case.
- **E010** [**BANK INPUT**] `docs/scenario_assumptions.md` – document; Text artifact with 5 line(s); Segment-level reasonableness review: Test whether scenario ordering remains directionally sensible at the segment level.; Methodology and documentation alignment review: Test whether written methodology, model overview, and overlay documentation match implemented behavior and observed outputs.
- **E011** [**BANK INPUT**] `model/cecl_engine.py` – code; Text artifact with 177 line(s); Package inventory and case classification: Determine whether the upload supports a model-driven CECL review and identify the core evidence types available.; Baseline reproduction: Confirm that the packaged baseline reserve can be reproduced from the supplied engine and portfolio inputs.; Portfolio-level scenario reruns: Run Baseline, Adverse, and Severe scenarios and assess directional reserve ordering.; Methodology and documentation alignment review: Test whether written methodology, model overview, and overlay documentation match implemented behavior and observed outputs.
- **E012** [**BANK INPUT**] `outputs/prior_baseline_reserve.csv` – packaged output table; CSV artifact with 1200 data row(s); Package inventory and case classification: Determine whether the upload supports a model-driven CECL review and identify the core evidence types available.; Baseline reproduction: Confirm that the packaged baseline reserve can be reproduced from the supplied engine and portfolio inputs.
- **E013** [**BANK INPUT**] `outputs/prior_scenario_summary.csv` – packaged output table; CSV artifact with 3 data row(s); Generated as part of the Codex review record for this case.
- **E014** [**BANK INPUT**] `outputs/prior_segment_reserves.csv` – packaged output table; CSV artifact with 12 data row(s); Generated as part of the Codex review record for this case.
- **E015** [**BANK INPUT**] `scenarios/adverse.csv` – scenario table; CSV artifact with 8 data row(s); Portfolio-level scenario reruns: Run Baseline, Adverse, and Severe scenarios and assess directional reserve ordering.

- **E016 [BANK INPUT]** `scenarios/baseline.csv` – scenario table; CSV artifact with 8 data row(s); Package inventory and case classification: Determine whether the upload supports a model-driven CECL review and identify the core evidence types available.; Portfolio-level scenario reruns: Run Baseline, Adverse, and Severe scenarios and assess directional reserve ordering.; Reserve driver bridge: Identify which macro and overlay drivers explain the majority of severe-versus-baseline reserve movement.
- **E017 [BANK INPUT]** `scenarios/severe.csv` – scenario table; CSV artifact with 8 data row(s); Portfolio-level scenario reruns: Run Baseline, Adverse, and Severe scenarios and assess directional reserve ordering.; Macro severity sensitivity: Test whether reserve responds proportionally to macro severity scaling.; Reserve driver bridge: Identify which macro and overlay drivers explain the majority of severe-versus-baseline reserve movement.

### Codex-Generated Review Artifacts

- **E018 [CODEX OUTPUT]** `outputs/support/discovery_summary.json` – json artifact; Text artifact with 81 line(s); Generated as part of the Codex review record for this case.
- **E019 [CODEX OUTPUT]** `outputs/support/discovery_summary.md` – document; Text artifact with 31 line(s); Summarizes what Codex identified in the uploaded package during discovery.
- **E020 [CODEX OUTPUT]** `outputs/support/case_understanding.md` – document; Text artifact with 25 line(s); Records Codex's synthesized view of the CECL process, assumptions, and constraints.
- **E021 [CODEX OUTPUT]** `outputs/support/review_plan.md` – document; Text artifact with 27 line(s); Captures the planned review procedures before detailed execution.
- **E022 [CODEX OUTPUT]** `outputs/support/review_strategy.md` – document; Text artifact with 81 line(s); Explains why specific procedures were selected from the discovered evidence.
- **E023 [CODEX OUTPUT]** `outputs/support/executed_test_matrix.md` – document; Text artifact with 14 line(s); Registers which procedures were executed, blocked, and what each procedure concluded.
- **E024 [CODEX OUTPUT]** `outputs/support/executed_test_matrix.csv` – generated review table; CSV artifact with 10 data row(s); Machine-readable form of the executed procedure register.
- **E025 [CODEX OUTPUT]** `outputs/support/evidence_map.md` – document; Text artifact with 73 line(s); Maps uploaded and derived artifacts to the procedures they supported.
- **E026 [CODEX OUTPUT]** `outputs/support/agentic_review_log.md` – document; Text artifact with 43 line(s); Readable stage-level summary of the intended Codex review flow.
- **E027 [CODEX OUTPUT]** `outputs/support/procedure_run_log.md` – document; Text artifact with 99 line(s); Chronological log of granular review actions and outputs.
- **E028 [CODEX OUTPUT]** `outputs/support/procedure_run_log.csv` – generated review table; CSV artifact with 12 data row(s); Machine-readable chronology of granular review actions.

- **E029 [CODEX OUTPUT]** `outputs/support/evidence_excerpts.md` – document; Text artifact with 190 line(s); Collects raw excerpts from uploaded materials and review outputs used in the writeup.
- **E030 [CODEX OUTPUT]** `outputs/support/baseline_reproduction.json` – json artifact; Text artifact with 37 line(s); Stores baseline reproduction metrics used in the full-review conclusion.
- **E031 [CODEX OUTPUT]** `outputs/support/scenario_results.csv` – generated review table; CSV artifact with 3 data row(s); Stores Codex-produced portfolio scenario rerun results.
- **E032 [CODEX OUTPUT]** `outputs/support/segment_reserve_comparison.csv` – generated review table; CSV artifact with 4 data row(s); Segment-level reasonableness review: Test whether scenario ordering remains directionally sensible at the segment level.
- **E033 [CODEX OUTPUT]** `outputs/support/sensitivity_results.csv` – generated review table; CSV artifact with 12 data row(s); Forecast horizon sensitivity: Assess reserve sensitivity to changing the reasonable-and-supportable forecast horizon.; Reversion speed sensitivity: Assess how severe-scenario reserve responds to longer reversion periods.; Macro severity sensitivity: Test whether reserve responds proportionally to macro severity scaling.; Overlay magnitude sensitivity: Assess the reserve effect of removing or amplifying qualitative overlays.
- **E034 [CODEX OUTPUT]** `outputs/support/driver_bridge.csv` – generated review table; CSV artifact with 6 data row(s); Reserve driver bridge: Identify which macro and overlay drivers explain the majority of severe-versus-baseline reserve movement.
- **E035 [CODEX OUTPUT]** `outputs/support/documentation_crosscheck.md` – document; Text artifact with 25 line(s); Stores documentation-versus-implementation and output cross-check observations.
- **E036 [CODEX OUTPUT]** `outputs/support/findings_register.json` – json artifact; Text artifact with 44 line(s); Registers findings, severity, and supporting evidence references.
- **E037 [CODEX OUTPUT]** `outputs/support/analysis_summary.json` – json artifact; Text artifact with 7 line(s); Generated as part of the Codex review record for this case.
- **E038 [CODEX OUTPUT]** `outputs/support/evidence_ledger.json` – json artifact; Text artifact with 364 line(s); Generated as part of the Codex review record for this case.
- **E039 [CODEX OUTPUT]** `outputs/support/coverage_statement.md` – document; Text artifact with 13 line(s); States supported versus blocked review coverage.
- **E040 [CODEX OUTPUT]** `outputs/support/codex_trace.md` – document; Text artifact with 33 line(s); Readable execution trace of discovery, planning, execution, and synthesis.

## Uploaded Package Tree

```
input_package/
  data/
    data_dictionary.csv
    loan_level_snapshot.csv
    overlay_schedule.csv
  docs/
```

```
    control_process_note.md
    governance_minutes.md
    methodology.md
    model_overview.md
    overlay_memo.md
    prior_review_note.md
    scenario_assumptions.md
  model/
    cecl_engine.py
  outputs/
    prior_baseline_reserve.csv
    prior_scenario_summary.csv
    prior_segment_reserves.csv
  scenarios/
    adverse.csv
    baseline.csv
    severe.csv
```

### Codex Output Tree

```
outputs/
  support/
    agentic_review_log.md
    analysis_summary.json
    baseline_reproduction.json
    case_understanding.md
    codex_trace.md
    coverage_statement.md
    discovery_summary.json
    discovery_summary.md
    documentation_crosscheck.md
    driver_bridge.csv
    evidence_excerpts.md
    evidence_ledger.json
    evidence_map.md
    executed_test_matrix.csv
    executed_test_matrix.md
    findings_register.json
    procedure_run_log.csv
    procedure_run_log.md
    review_plan.md
    review_strategy.md
    scenario_results.csv
    segment_reserve_comparison.csv
    sensitivity_results.csv
```

## Package Discovery and Case Understanding

The discovered package is best understood as a model-driven CECL review case for Redwood Regional Bank's Q1 2026 CECL Allowance Review. It includes executable reserve logic, loan-level

portfolio data, scenario definitions, prior baseline outputs, and enough supporting documentation to support both quantitative reruns and documentation challenge.

- Central assumption or theme: Reasonable-and-supportable horizon is documented as 4 quarters, with 4 quarters of reversion.
- Central assumption or theme: Scenario severity should increase reserve directionally from Baseline to Adverse to Severe at portfolio level and generally by segment.
- Central assumption or theme: Qualitative overlays should remain within documented governance guardrails and should not produce unexplained segment relief.
- Reviewable scope: Baseline reproduction from supplied engine and data.
- Reviewable scope: Scenario reruns and segment-level reasonableness testing.
- Reviewable scope: Sensitivity testing for horizon, reversion, macro severity, and overlays.
- Reviewable scope: Methodology, scenario, and overlay documentation alignment review.
- Key risk to challenge: Documentation may not match the implemented forecast and reversion treatment.
- Key risk to challenge: Overlay governance may not reflect actual overlay usage by scenario and segment.
- Key risk to challenge: A severe scenario can appear directionally harsher at portfolio level while still producing a segment anomaly.
- Constraint: No independent historical performance dataset or external benchmark was supplied; conclusions are bounded to the package and synthetic portfolio.

## CECL Process Understanding

The package describes a loan-level CECL process spanning 4 segments: Commercial Real Estate, Commercial and Industrial, Residential Mortgage, Consumer Unsecured. The quantitative framework applies scenario-dependent macro paths to lifetime PD and LGD components and then layers qualitative overlays by segment and scenario.

| Assumption Area | Documented Position | Observed in Package | Review Implication |
|---|---|---|---|
| Forecast horizon | 4 quarters documented | 6 quarters implemented | Methodology and model overview are not aligned to engine behavior. |
| Reversion horizon | 4 quarters documented | 2 quarters implemented | Policy description and actual reserve mechanics differ. |
| Overlay cap | 5.0 bps documented | 9.0 bps scheduled | Qualitative adjustment magnitude exceeds documented guardrail. |
| Scenario reasonableness | Severe should be directionally harsher unless clearly justified | Residential Mortgage reserve falls by $109,988 from adverse to severe | Segment-level severe path requires additional governance challenge. |

## Review Strategy and Planning Logic

The review strategy was formed from the discovered evidence. Codex considered 5 central review questions, selected 10 executable procedures, and identified 0 blocked or non-applicable procedures. Because the package contained runnable code, scenario files, prior outputs, and core methodology documents, the review strategy prioritized execution-based challenge first and then used documentation review to test whether the written methodology and governance artifacts actually described the observed reserve behavior. The table below is the high-level procedure register. It is meant to communicate what Codex was trying to prove before running the work. The more detailed planning record remains in the support artifacts listed immediately after the table.

1. Does the package contain enough evidence to support a full execution-based CECL review rather than a documentation-only assessment?
2. Can the supplied baseline reserve be reproduced from the packaged reserve engine, portfolio data, and baseline scenario?
3. Do reserve outputs move directionally and materially across Baseline, Adverse, and Severe scenarios both overall and by major segment?
4. How sensitive are reserves to forecast horizon, reversion speed, macro severity, and overlay magnitude assumptions?
5. Do the methodology, scenario, and overlay documents faithfully describe the behavior observed in the implemented CECL process, especially for Residential Mortgage?

| ID | Procedure | What Codex Was Trying To Test | Status |
|---|---|---|---|
| FR-01 | Package inventory and case classification | Determine whether the upload supports a model-driven CECL review and identify the core evidence types available. | Executed |
| FR-02 | Baseline reproduction | Confirm that the packaged baseline reserve can be reproduced from the supplied engine and portfolio inputs. | Executed |
| FR-03 | Portfolio-level scenario reruns | Run Baseline, Adverse, and Severe scenarios and assess directional reserve ordering. | Executed |
| FR-04 | Segment-level reasonableness review | Test whether scenario ordering remains directionally sensible at the segment level. | Executed |
| FR-05 | Forecast horizon sensitivity | Assess reserve sensitivity to changing the reasonable-and-supportable forecast horizon. | Executed |
| FR-06 | Reversion speed sensitivity | Assess how severe-scenario reserve responds to longer reversion periods. | Executed |
| FR-07 | Macro severity sensitivity | Test whether reserve responds proportionally to macro severity scaling. | Executed |
| FR-08 | Overlay magnitude sensitivity | Assess the reserve effect of removing or amplifying qualitative overlays. | Executed |
| FR-09 | Reserve driver bridge | Identify which macro and overlay drivers explain the majority of severe-versus-baseline reserve movement. | Executed |
| FR-10 | Methodology and documentation alignment review | Test whether written methodology, model overview, and overlay documentation match implemented behavior and observed outputs. | Executed |

### Planning and Execution Record Files

The following Codex-generated files contain the full planning rationale, executed procedure register, evidence mapping, and granular step log for this review.

- `outputs/support/review_plan.md`
- `outputs/support/review_strategy.md`
- `outputs/support/executed_test_matrix.md`
- `outputs/support/evidence_map.md`
- `outputs/support/procedure_run_log.md`

## Selected Raw Evidence Excerpts

The excerpts below are short verbatim snippets from bank-uploaded materials and Codex-generated outputs that support the main findings and illustrate the review lineage.

### [BANK INPUT] docs/methodology.md

Documented CECL forecast and reversion assumptions cited in the horizon mismatch finding.

```
# CECL Methodology

## Objective
Review the Q1 2026 CECL Allowance Review CECL reserve process for Redwood Regional Bank. The documented

## Segments
- Commercial Real Estate
- Commercial and Industrial
- Residential Mortgage
- Consumer Unsecured

## Key Drivers
- Unemployment rate
- GDP growth
- House price growth
- CRE price growth
- Prime rate

## Overlay Treatment
Qualitative overlays are documented as targeted adjustments capped at 5.0 bps and are not expected to cl
```

### [BANK INPUT] model/cecl_engine.py

Implemented reserve-engine constants and overlay schedule that Codex compared to the methodology and overlay memo.

```
"cre_price_growth": 0.0,
      "prime_rate": 0.11
    }
  }
}
OVERLAYS = {
  "baseline": {
    "commercial_real_estate": 2.0,
    "commercial_and_industrial": 2.0,
    "residential_mortgage": 1.0,
    "consumer_unsecured": 3.0
  },
  "adverse": {
    "commercial_real_estate": 5.0,
    "commercial_and_industrial": 4.0,
    "residential_mortgage": 2.0,
    "consumer_unsecured": 6.0
  },
  "severe": {
    "commercial_real_estate": 8.0,
    "commercial_and_industrial": 7.0,
    "residential_mortgage": -9.0,
    "consumer_unsecured": 9.0
  }
}
FORECAST_QUARTERS = 6
REVERSION_QUARTERS = 2


def build_projection(frame: pd.DataFrame) -> pd.DataFrame:
    long_run = {
        "unemployment_rate": 4.7,
        "gdp_growth": 1.7,
        "house_price_growth": 1.9,
        "cre_price_growth": 1.4,
        "prime_rate": 5.0,
    }
    forecast = frame.iloc[:FORECAST_QUARTERS].copy()
    last = forecast.iloc[-1].copy()
    reversion_rows = []
    for step in range(1, REVERSION_QUARTERS + 1):
        weight = step / REVERSION_QUARTERS
        row = {"quarter": f"reversion_{step}"}
        for key, long_run_value in long_run.items():
            row[key] = round(float(last[key]) + (long_run_value - float(last[key])) * weight, 4)
        reversion_rows.append(row)
```

## [BANK INPUT] docs/overlay_memo.md

Documented overlay cap and severe overlay posture used in the overlay-governance challenge.

```
# Overlay Memo
```

```
Management overlays are described as targeted and modest, capped at 5.0 bps by segment-scenario combina
```

## [BANK INPUT] scenarios/adverse.csv + scenarios/severe.csv

Numeric adverse and severe scenario paths that Codex compared for directional severity and segment reasonableness.

```
[adverse.csv]
quarter,unemployment_rate,gdp_growth,house_price_growth,cre_price_growth,prime_rate
2026Q1,5.4,0.8,-1.5,-2.2,5.6
2026Q2,5.9,0.3,-3.2,-4.1,5.75
2026Q3,6.3,-0.4,-4.8,-5.6,5.75
2026Q4,6.1,-0.1,-4.4,-4.7,5.55
2027Q1,5.8,0.2,-2.8,-3.1,5.4
2027Q2,5.5,0.6,-1.3,-1.5,5.25
2027Q3,5.1,1.0,0.1,-0.4,5.1
2027Q4,4.9,1.2,0.8,0.2,5.0

[severe.csv]
quarter,unemployment_rate,gdp_growth,house_price_growth,cre_price_growth,prime_rate
2026Q1,6.0,0.2,-2.0,-3.2,5.75
2026Q2,6.9,-0.8,-4.4,-6.6,6.0
2026Q3,7.6,-1.6,-5.8,-8.0,6.05
2026Q4,7.2,-0.9,-2.9,-5.1,5.85
2027Q1,6.6,-0.2,0.4,-2.6,5.55
2027Q2,6.0,0.4,1.6,-1.0,5.3
2027Q3,5.4,0.9,1.8,0.1,5.1
2027Q4,5.0,1.2,2.0,0.6,5.0
```

## [CODEX OUTPUT] outputs/support/baseline_reproduction.json

Codex-generated reproduction record showing the packaged baseline reserve and the rerun reserve matched exactly.


```
{
  "packaged_total_reserve": 4876841.47,
  "rerun_total_reserve": 4876841.47,
  "absolute_delta": 0.0,
  "pct_delta": 0.0,
  "max_loan_level_abs_delta": 0.0,
  "segment_deltas": [
    {
      "segment_id": "commercial_and_industrial",
      "reserve_amount_packaged": 1977246.19,
      "reserve_amount_rerun": 1977246.19,
      "delta": 0.0,
      "pct_delta": 0.0
    },
    {
      "segment_id": "commercial_real_estate",
      "reserve_amount_packaged": 2545519.51,
```

```
      "reserve_amount_rerun": 2545519.51,
      "delta": 0.0,
      "pct_delta": 0.0
    },
    {
      "segment_id": "consumer_unsecured",
      "reserve_amount_packaged": 100355.66,
      "reserve_amount_rerun": 100355.66,
      "delta": 0.0,
      "pct_delta": 0.0
    },
    {
      "segment_id": "residential_mortgage",
      "reserve_amount_packaged": 253720.11,
      "reserve_amount_rerun": 253720.11,
      "delta": 0.0,
      "pct_delta": 0.0
    }
  ]
}
```


## [CODEX OUTPUT] outputs/support/segment_reserve_comparison.csv

Codex-generated segment comparison highlighting the Residential Mortgage severe-versus-adverse anomaly.


```
segment_id,adverse,baseline,severe,adverse_minus_baseline,severe_minus_adverse
commercial_and_industrial,2712808.82,1977246.19,3148323.76,735562.63,435514.94
commercial_real_estate,3972723.25,2545519.51,4787769.99,1427203.74,815046.74
consumer_unsecured,118790.69,100355.66,133898.81,18435.03,15108.12
residential_mortgage,327699.21,253720.11,217710.91,73979.1,-109988.3
```


## [CODEX OUTPUT] outputs/support/sensitivity_results.csv

Codex-generated sensitivity results used to challenge forecast horizon, reversion, macro severity, and overlay magnitude assumptions.


```
test_type,setting,scenario_name,reserve_amount
forecast_horizon,4,baseline,4877986.54
forecast_horizon,6,baseline,4876841.47
forecast_horizon,8,baseline,4876841.47
reversion_speed,2,severe,8287703.47
reversion_speed,4,severe,8384330.79
reversion_speed,6,severe,8472904.31
macro_severity_scale,0.75,severe,7126890.49
macro_severity_scale,1.0,severe,8287703.47
macro_severity_scale,1.25,severe,9803105.55
overlay_multiplier,0.0,severe,7850916.88
overlay_multiplier,1.0,severe,8287703.47
overlay_multiplier,1.5,severe,8506096.88
```

## Procedures Performed

### FR-01 - Package inventory and case classification

Determine whether the upload supports a model-driven CECL review and identify the core evidence types available.

- **Status:** executed
- **Why selected:** Runnable code, scenario tables, loan-level data, prior outputs, and methodology documents were all present.
- **Bank-input evidence relied upon:** `model/cecl_engine.py`, `data/loan_level_snapshot.csv`, `scenarios/baseline.csv`, `outputs/prior_baseline_reserve.csv`, `docs/methodology.md`
- **Codex-generated evidence relied upon:** *None*
- **Codex-generated outputs produced:** `discovery_summary.md`, `evidence_ledger.json`
- **Assessment criteria:** Review should only proceed as execution-based if engine, data, scenarios, and prior outputs are all available.
- **Key result:** Package classified as model-driven CECL review with no material execution blockers.

### FR-02 - Baseline reproduction

Confirm that the packaged baseline reserve can be reproduced from the supplied engine and portfolio inputs.

- **Status:** executed
- **Why selected:** A packaged prior baseline reserve was supplied alongside the runnable engine and source portfolio.
- **Bank-input evidence relied upon:** `model/cecl_engine.py`, `data/loan_level_snapshot.csv`, `outputs/prior_baseline_reserve.csv`
- **Codex-generated evidence relied upon:** *None*
- **Codex-generated outputs produced:** `baseline_reproduction.json`
- **Assessment criteria:** Total reserve and segment reserve should reconcile within tolerance; large deltas would undermine confidence in package integrity.
- **Key result:** Baseline rerun matched packaged reserve exactly at 4876841.47 with 0.000% delta.

### FR-03 - Portfolio-level scenario reruns

Run Baseline, Adverse, and Severe scenarios and assess directional reserve ordering.

- **Status:** executed

- **Why selected:** The package included three explicit scenario files and a model designed to consume them.
- **Bank-input evidence relied upon:** `scenarios/baseline.csv`, `scenarios/adverse.csv`, `scenarios/severe.csv`, `model/cecl_engine.py`
- **Codex-generated evidence relied upon:** *None*
- **Codex-generated outputs produced:** `scenario_results.csv`
- **Assessment criteria:** Portfolio reserve should generally increase with scenario severity absent a documented exception.
- **Key result:** Portfolio reserve increased from 4876841.47 to 7132021.97 to 8287703.47.

### FR-04 - Segment-level reasonableness review

Test whether scenario ordering remains directionally sensible at the segment level.

- **Status:** executed
- **Why selected:** CECL governance depends on segment-level reserve understanding, not just portfolio totals.
- **Bank-input evidence relied upon:** `docs/scenario_assumptions.md`, `docs/overlay_memo.md`
- **Codex-generated evidence relied upon:** `outputs/support/segment_reserve_comparison.csv`
- **Codex-generated outputs produced:** `segment_reserve_comparison.csv`
- **Assessment criteria:** Segment movements should generally align to scenario severity or be explicitly supported by documented assumptions.
- **Key result:** Residential Mortgage severe reserve was 217710.91, below adverse reserve 327699.21, creating a documented challenge item.

### FR-05 - Forecast horizon sensitivity

Assess reserve sensitivity to changing the reasonable-and-supportable forecast horizon.

- **Status:** executed
- **Why selected:** The methodology makes forecast horizon a named governance assumption.
- **Bank-input evidence relied upon:** `docs/methodology.md`
- **Codex-generated evidence relied upon:** `outputs/support/sensitivity_results.csv`
- **Codex-generated outputs produced:** `sensitivity_results.csv`
- **Assessment criteria:** Reserve movement should be understandable and documented; mismatch to stated policy must be surfaced.
- **Key result:** Moving from 6 to 4 forecast quarters changed baseline reserve only modestly, but it directly exposed a documented-versus-implemented policy mismatch.

### FR-06 - Reversion speed sensitivity

Assess how severe-scenario reserve responds to longer reversion periods.

- **Status:** executed
- **Why selected:** Reversion mechanics are central to CECL governance and are explicitly documented in the package.
- **Bank-input evidence relied upon:** `docs/methodology.md`
- **Codex-generated evidence relied upon:** `outputs/support/sensitivity_results.csv`
- **Codex-generated outputs produced:** `sensitivity_results.csv`
- **Assessment criteria:** Reserve should move directionally as reversion assumptions are extended or shortened.
- **Key result:** Extending severe-scenario reversion from 2 to 6 quarters increased reserve materially, confirming reversion is a meaningful reserve lever.

### FR-07 - Macro severity sensitivity

Test whether reserve responds proportionally to macro severity scaling.

- **Status:** executed
- **Why selected:** Scenario severity and directional reasonableness are core to CECL challenge work.
- **Bank-input evidence relied upon:** `scenarios/severe.csv`
- **Codex-generated evidence relied upon:** `outputs/support/sensitivity_results.csv`
- **Codex-generated outputs produced:** `sensitivity_results.csv`
- **Assessment criteria:** Scaling macro severity upward should generally increase reserve; weaker response would require explanation.
- **Key result:** Scaling severe stress from 1.00x to 1.25x materially increased reserve, supporting macro sensitivity monotonicity at portfolio level.

### FR-08 - Overlay magnitude sensitivity

Assess the reserve effect of removing or amplifying qualitative overlays.

- **Status:** executed
- **Why selected:** The package included a scenario-by-segment overlay schedule and an overlay governance memo.
- **Bank-input evidence relied upon:** `data/overlay_schedule.csv`, `docs/overlay_memo.md`
- **Codex-generated evidence relied upon:** `outputs/support/sensitivity_results.csv`

- **Codex-generated outputs produced:** `sensitivity_results.csv`
- **Assessment criteria:** Overlay effect should be quantitatively explainable and consistent with governance statements.
- **Key result:** Removing severe overlays reduced reserve materially, confirming overlay posture is consequential and governance-relevant.

### FR-09 - Reserve driver bridge

Identify which macro and overlay drivers explain the majority of severe-versus-baseline reserve movement.

- **Status:** executed
- **Why selected:** A driver bridge makes the review more interpretable and supports reasonableness challenge.
- **Bank-input evidence relied upon:** `scenarios/baseline.csv`, `scenarios/severe.csv`
- **Codex-generated evidence relied upon:** `outputs/support/driver_bridge.csv`
- **Codex-generated outputs produced:** `driver_bridge.csv`
- **Assessment criteria:** Major reserve movement should be attributable to understandable drivers rather than unexplained residual behavior.
- **Key result:** CRE price growth was the largest modeled standalone driver of severe-versus-baseline reserve change.

### FR-10 - Methodology and documentation alignment review

Test whether written methodology, model overview, and overlay documentation match implemented behavior and observed outputs.

- **Status:** executed
- **Why selected:** The package contained enough documentation to support direct cross-checking against the model and results.
- **Bank-input evidence relied upon:** `docs/methodology.md`, `docs/model_overview.md`, `docs/scenario_ass` `docs/overlay_memo.md`, `model/cecl_engine.py`
- **Codex-generated evidence relied upon:** *None*
- **Codex-generated outputs produced:** `documentation_crosscheck.md`, `findings_register.json`
- **Assessment criteria:** Documented horizon, reversion, scenario logic, and overlay posture should match the implemented process or be clearly qualified.
- **Key result:** Cross-checking identified a forecast/reversion mismatch, overlay cap mismatch, and a severe-scenario segment anomaly requiring remediation.

## Agentic Execution Workflow

### Discovery

The review started by inventorying the package and determining whether a quantitative review was actually possible from the supplied materials.

- Review question addressed: What is in the package and does it support execution-based CECL review?
- Decision or rationale: Treat the case as model-driven because reserve code, loan-level data, scenarios, prior outputs, and core methodology documents were all present.
- Inputs reviewed: input_package/*
- Outputs produced: discovery_summary.md, evidence_ledger.json, case_understanding.md

### Case understanding and planning

Codex formed a review strategy around the strongest available evidence and explicitly selected a test set before running the work.

- Review question addressed: Which review questions matter most for this package and which procedures can answer them credibly?
- Decision or rationale: Prioritize baseline reproduction, scenario reruns, sensitivities, and documentation alignment because the package supports all four lanes.
- Inputs reviewed: discovery_summary.md, docs/methodology.md, docs/scenario_assumptions.md, docs/overlay_memo.md
- Outputs produced: review_plan.md, review_strategy.md, executed_test_matrix.md

### Quantitative execution

The review ran the CECL engine against baseline and stressed scenarios, then executed targeted sensitivities to challenge the major reserve assumptions.

- Review question addressed: Can the reserve process be reproduced, and do outputs move sensibly overall and by segment?
- Decision or rationale: Run baseline, adverse, and severe scenarios first, then test horizon, reversion, macro severity, and overlays as the most consequential levers.
- Inputs reviewed: model/cecl_engine.py, data/loan_level_snapshot.csv, scenarios/*.csv, data/overlay_schedule
- Outputs produced: baseline_reproduction.json, scenario_results.csv, segment_reserve_comparison.csv, sensitivity_results.csv, driver_bridge.csv

### Documentation challenge

After quantitative outputs were produced, the review cross-checked the documents against implemented behavior and reserve results.

- Review question addressed: Do the written methodology, scenario, and overlay materials describe the same process that the results imply?
- Decision or rationale: Escalate mismatches where documented governance assumptions differ from implementation or where narrative support is insufficient for observed output behavior.
- Inputs reviewed: docs/*.md, model/cecl_engine.py, segment_reserve_comparison.csv, sensitivity_results.csv
- Outputs produced: documentation_crosscheck.md, findings_register.json, evidence_map.md

### Synthesis

The final stage combined discovery, planning, quantitative results, and documentation challenge into an internal review report and support pack.

- Review question addressed: What opinion can be supported, what findings should be issued, and what evidence supports that conclusion?
- Decision or rationale: Issue a substantive review with explicit remediation for documentation mismatch, overlay governance, and the residential mortgage severe-scenario anomaly.
- Inputs reviewed: findings_register.json, executed_test_matrix.md, coverage_statement.md, agentic_review_log.md
- Outputs produced: cecl_review_memo.tex, cecl_review_memo.pdf

## Chronological Procedure and Output Log

This section records the more granular action sequence for the review, including the specific bank inputs examined, the Codex-generated outputs created, and the observed result at each step.

### EVT-01 - Inventory uploaded package

- **Phase:** discovery
- **Related procedure:** FR-01
- **Action:** Enumerated the input package and confirmed presence of reserve engine, loan-level data, scenarios, prior outputs, and CECL methodology documents.
- **Inputs:** `input_package/model/cecl_engine.py`, `input_package/data/loan_level_snapshot.csv`, `input_package/scenarios/*.csv`, `input_package/docs/*.md`
- **Outputs:** `outputs/support/discovery_summary.md`, `outputs/support/evidence_ledger.json`
- **Result:** Case classified as execution-capable CECL review with no material runtime blockers.

### EVT-02 - Select review lanes

- **Phase:** planning
- **Related procedure:** FR-01
- **Action:** Chose baseline reproduction, scenario reruns, sensitivity testing, reserve bridge, and documentation challenge as the primary review lanes.
- **Inputs:** `outputs/support/discovery_summary.md`, `input_package/docs/methodology.md`, `input_package/docs/overlay_memo.md`
- **Outputs:** `outputs/support/review_plan.md`, `outputs/support/review_strategy.md`
- **Result:** Ten executable procedures selected; none blocked.

### EVT-03 - Reproduce packaged baseline reserve

- **Phase:** execution
- **Related procedure:** FR-02
- **Action:** Reran the packaged engine under the baseline scenario and reconciled the reproduced reserve to the supplied baseline reserve file.
- **Inputs:** `input_package/model/cecl_engine.py`, `input_package/data/loan_level_snapshot.csv`, `input_package/scenarios/baseline.csv`, `input_package/outputs/prior_baseline_reserve.csv`
- **Outputs:** `outputs/support/baseline_reproduction.json`
- **Result:** Packaged reserve 4876841.47; rerun reserve 4876841.47; absolute delta 0.00.

### EVT-04 - Run baseline, adverse, and severe scenarios

- **Phase:** execution
- **Related procedure:** FR-03
- **Action:** Executed the reserve engine across the three supplied scenario paths and aggregated total reserve results.
- **Inputs:** `input_package/model/cecl_engine.py`, `input_package/data/loan_level_snapshot.csv`, `input_package/scenarios/baseline.csv`, `input_package/scenarios/adverse.csv`, `input_package/scena
- **Outputs:** `outputs/support/scenario_results.csv`
- **Result:** Portfolio reserve moved from 4876841.47 to 7132021.97 to 8287703.47.

### EVT-05 - Check segment-level scenario ordering

- **Phase:** execution
- **Related procedure:** FR-04

- **Action:** Compared segment reserve amounts across baseline, adverse, and severe scenarios to test directional reasonableness by segment.
- **Inputs:** `outputs/support/scenario_results.csv`, `input_package/docs/scenario_assumptions.md`, `input_package/docs/overlay_memo.md`
- **Outputs:** `outputs/support/segment_reserve_comparison.csv`
- **Result:** Residential Mortgage reserve fell from 327699.21 in adverse to 217710.91 in severe.

### EVT-06 - Run forecast-horizon sensitivity

- **Phase:** execution
- **Related procedure:** FR-05
- **Action:** Tested baseline reserve under alternate forecast horizons to challenge the documented 4-quarter assumption versus the 6-quarter implementation.
- **Inputs:** `input_package/docs/methodology.md`, `input_package/model/cecl_engine.py`
- **Outputs:** `outputs/support/sensitivity_results.csv`
- **Result:** Changing the baseline forecast horizon from 6 quarters to 4 quarters changed reserve modestly, but confirmed a policy-versus-implementation mismatch.

### EVT-07 - Run reversion-speed sensitivity

- **Phase:** execution
- **Related procedure:** FR-06
- **Action:** Re-estimated severe-scenario reserve using longer reversion periods to test sensitivity to the reversion assumption.
- **Inputs:** `input_package/docs/methodology.md`, `input_package/model/cecl_engine.py`
- **Outputs:** `outputs/support/sensitivity_results.csv`
- **Result:** Extending severe reversion from 2 to 6 quarters increased reserve from 8287703.47 to 8472904.31.

### EVT-08 - Run macro-severity scaling sensitivity

- **Phase:** execution
- **Related procedure:** FR-07
- **Action:** Scaled severe macro stress to test whether reserve responds monotonically to harsher macro conditions.
- **Inputs:** `input_package/scenarios/severe.csv`, `input_package/model/cecl_engine.py`
- **Outputs:** `outputs/support/sensitivity_results.csv`

- **Result:** Increasing severe macro scaling from 1.00x to 1.25x increased reserve from 8287703.47 to 9803105.55.

### EVT-09 - Run overlay-magnitude sensitivity

- **Phase:** execution
- **Related procedure:** FR-08
- **Action:** Removed and amplified overlays to measure how much of the severe reserve outcome is attributable to overlay posture.
- **Inputs:** `input_package/data/overlay_schedule.csv`, `input_package/docs/overlay_memo.md`
- **Outputs:** `outputs/support/sensitivity_results.csv`
- **Result:** Removing severe overlays reduced reserve from 8287703.47 to 7850916.88; amplifying to 1.50x raised reserve to 8506096.88.

### EVT-10 - Build reserve-driver bridge

- **Phase:** execution
- **Related procedure:** FR-09
- **Action:** Constructed a simple driver bridge to attribute severe-versus-baseline reserve movement to major macro and overlay factors.
- **Inputs:** `input_package/scenarios/baseline.csv`, `input_package/scenarios/severe.csv`, `input_package/model/cecl_engine.py`
- **Outputs:** `outputs/support/driver_bridge.csv`
- **Result:** CRE price growth was the largest standalone driver, contributing 1108566.01 of severe-versus-baseline reserve change.

### EVT-11 - Cross-check methodology against implementation

- **Phase:** documentation_challenge
- **Related procedure:** FR-10
- **Action:** Compared documented forecast, reversion, scenario, and overlay assumptions to the implemented engine constants and quantitative outputs.
- **Inputs:** `input_package/docs/methodology.md`, `input_package/docs/model_overview.md`, `input_package/docs/overlay_memo.md`, `input_package/model/cecl_engine.py`, `outputs/support/segmen`
- **Outputs:** `outputs/support/documentation_crosscheck.md`, `outputs/support/findings_register.json`
- **Result:** Recorded horizon/reversion mismatch, overlay-cap mismatch, and severe Residential Mortgage anomaly as formal findings.

### EVT-12 - Assemble internal review pack

- **Phase:** synthesis
- **Related procedure:** FR-10
- **Action:** Combined discovery, planning, quantitative outputs, documentation challenge, and evidence mapping into the final internal-review report pack.
- **Inputs:** `outputs/support/review_strategy.md`, `outputs/support/procedure_run_log.md`, `outputs/support/findings_register.json`, `outputs/support/evidence_map.md`
- **Outputs:** `outputs/stakeholder/cecl_review_memo.tex`, `outputs/stakeholder/cecl_review_memo.pdf`
- **Result:** Produced a comprehensive internal CECL review memo with explicit provenance, raw evidence excerpts, and procedure-level detail.

## Quantitative Results

### Baseline Reproduction

Baseline reproduction landed at $4,876,841 against a packaged baseline of $4,876,841, with absolute delta $0 and maximum loan-level delta $0. Relative to Baseline, Adverse increased reserve by $2,255,180 and Severe increased reserve by another $1,155,682, for a total Severe-versus-Baseline increase of $3,410,862.

| Metric | Result |
|---|---|
| Packaged baseline reserve | $4,876,841 |
| Rerun baseline reserve | $4,876,841 |
| Absolute delta | $0 |
| Percent delta | 0.000% |
| Maximum loan-level absolute delta | $0 |

### Scenario Results

Portfolio-level reserve ordering was monotonic. Adverse increased reserve by $2,255,180 versus Baseline, and Severe increased reserve by $1,155,682 versus Adverse. Detailed scenario-level reserve values are recorded in outputs/support/scenario_results.csv. That file should be treated as the full machine-readable record for portfolio scenario reruns.

### Segment Results

Segment-level behavior was directionally reasonable for most segments. The notable exception was Residential Mortgage, which declined by $109,988 from Adverse to Severe. The segment-by-segment reserve record is maintained in outputs/support/segment_reserve_comparison.csv. That file should be used for full segment reconciliation and for tracing the Residential Mortgage anomaly.

### Sensitivity Testing

Sensitivity testing showed limited change from moving the documented forecast horizon to 4 quarters (change of $1,145), but reserve meaningfully increased as reversion lengthened from 2 to 6 quarters under Severe (change of $185,201). Scaling severe macro stress from 1.00x to 1.25x added $1,515,402, while removing overlays reduced Severe reserve by $436,787.

- Forecast horizon: moving from 6 to 4 forecast quarters changed reserve by $1,145.
- Reversion speed: extending severe-scenario reversion from 2 to 6 quarters increased reserve by $185,201.
- Macro severity: scaling severe stress from 1.00x to 1.25x increased reserve by $1,515,402.
- Overlay magnitude: removing overlays reduced reserve by $436,787.

The complete sensitivity grid, including all tested settings and reserve outputs, is recorded in outputs/support/sensitivity_results.csv.

### Driver Analysis

The simple reserve bridge indicates that Cre Price Growth was the largest modeled contributor to the Severe-versus-Baseline reserve increase, adding $1,108,566 on a standalone basis. The full reserve-driver bridge is recorded in outputs/support/driver_bridge.csv and should be used for any deeper driver reconciliation or appendix work.

### Quantitative Support Files

- `outputs/support/baseline_reproduction.json`
- `outputs/support/scenario_results.csv`
- `outputs/support/segment_reserve_comparison.csv`
- `outputs/support/sensitivity_results.csv`
- `outputs/support/driver_bridge.csv`

## Documentation and Assumption Alignment

Documentation cross-checking found that the written methodology still describes a 4-quarter reasonable-and-supportable period with 4-quarter reversion, while the engine implements 6 forecast quarters and 2 reversion quarters. The overlay memo references a 5.0 bps guardrail, but the supplied overlay schedule applies up to 9.0 bps. At segment level, Residential Mortgage shows a severe-scenario reserve $217,711, below its adverse reserve $327,699, driven by the scenario and overlay posture described in the package.

- Documented forecast/reversion: 4Q forecast and 4Q reversion.
- Implemented forecast/reversion: 6Q forecast and 2Q reversion.

- Documented overlay cap: 5.0 bps; scheduled overlay cap: 9.0 bps.
- Anomaly observation: Residential Mortgage severe reserve $217,711 versus adverse reserve $327,699.

### Documentation Review Files

- `outputs/support/documentation_crosscheck.md`
- `outputs/support/findings_register.json`
- `outputs/support/evidence_excerpts.md`

## Detailed Findings and Remediation

### Documented CECL horizon and reversion do not match the implemented reserve engine

**Severity:** HIGH
**Observation:** Documentation describes a 4-quarter reasonable-and-supportable period with 4-quarter reversion, while the reserve engine is configured for 6 forecast quarters and 2 reversion quarters.
**Evidence:** docs/methodology.md, docs/model_overview.md, model/cecl_engine.py
**Recommended response:** Either update the methodology and model overview to the implemented 6-quarter forecast and 2-quarter reversion, or change the engine to match documented policy before the next governance submission.

### Qualitative overlay effect is materially larger than documentation suggests

**Severity:** MEDIUM
**Observation:** The documentation frames qualitative overlays as capped at 5.0 bps, but the supplied overlay table uses up to 9.0 bps.
**Evidence:** docs/overlay_memo.md, data/overlay_schedule.csv, outputs/support/sensitivity_results.csv
**Recommended response:** Reconcile the overlay memo and governance materials to the actual schedule, including the 9.0 bps cap and the rationale for segment-level relief under Severe.

### One segment behaves directionally oddly under the severe scenario

**Severity:** HIGH
**Observation:** Segment residential_mortgage produces a severe-scenario reserve of 217710.91, below the adverse reserve of 327699.21. The severe path allows house prices to recover faster than adverse after 2026Q4 while the overlay schedule applies negative mortgage relief, creating a directionally odd segment result.
**Evidence:** outputs/support/segment_reserve_comparison.csv, scenarios/severe.csv, docs/scenario_assumptions.
docs/overlay_memo.md
**Recommended response:** Challenge the Residential Mortgage severe scenario result, document

whether the house-price recovery and negative overlay are intended, and update committee materials accordingly.

### Baseline reserve was reproducible within tolerance

**Severity:** LOW
**Observation:** Rerunning the supplied baseline logic reproduced the packaged baseline reserve within 0.000%.
**Evidence:** outputs/prior_baseline_reserve.csv, outputs/support/baseline_reproduction.json
**Recommended response:** Retain the reproduction evidence as support for package completeness and execution readiness.

## Overall Assessment

The supplied package supported a substantive CECL review and allowed execution of the core quantitative and documentation work expected for this scope. The baseline output was reproducible and overall reserve direction was reasonable, but documentation and governance artifacts are not yet aligned to implemented behavior, and one segment-level severe result requires explicit management challenge before the package would support an unqualified review conclusion.